

An Efficient NSKDP Authentication Method to Secure Smart Grid

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Abstract—Since the Information Networks are added to the current electricity networks, the security and privacy of individuals is challenged. This combination of technologies creates vulnerabilities in the context of smart grid power which disrupt the consumer energy supply. Methods based on encryption are against the countermeasures attacks that have targeted the integrity and confidentiality factors. Although the cryptography strategies are used in Smart Grid, key management which is different in size from tens to millions of keys (for meters), is considered as the critical processes.

The Key mismanagement causes to reveal the secret keys for attacker, a symmetric key distribution method is recently suggested by [7] which is based on a symmetric key distribution, this strategy is very suitable for smart electric meters. The problem with this method is its vulnerability to impersonating respondents attack. The proposed approach to solve this problem is to send the both side identifiers in encrypted form based on hash functions and a random value, the proposed solution is appropriate for devices such as meters that have very little computing power.

Keywords—vulnerability; cyber security; cryptographic protocols; smart meters; smart grid

I. INTRODUCTION

Environmental and economic factors have caused the old electrical infrastructure become unstable. Thus, a demand for reliability, compatibility, security and cost-effectiveness was requested. Electrical infrastructure with these multiple features is called Smart Grid systems. This smart systems must be used for higher productivity and lower consumption. There are several definitions for Smart Grid for example U.S. Department of Energy has given a definition which is as follows:

As shown in Fig. 1, the lower infrastructure of physical layer is for energy distribution. Communication infrastructure is placed at the top of energy infrastructure and create and manage all the chain communications across the network. The third layer, is a computing layer for decision-making in real-time situations and is placed at the top of the applications and smart devices for making the required data. As you can see, security is not related to a specific layer and involve all of them.

Analysis and implementation of Smart Grid is a challenging task, especially when the amount of potential damages that could be caused by cyber-attacks is considered,

the need to protect the security of elements in the smart grid data is arisen from the interaction between smart grid with a two-way communication network that enable energy suppliers and consumers share information with each other through interactive and real situations [1]-[3]. To ensure the operation to be safe and reliable, before a comprehensive overview of network security is mentioned, the security objectives and security requirements need to be explained.

II. THE SECURITY OBJECTIVES AND REQUERMENTS OF THE SMART GRID

Network security coordination in the Council Working Group has released a comprehensive policy on the Smart Grid security. According to The report, security purposes in Smart Grid are as below:

- **Reliability:** Ensuring reliable and timely access to information is the most important thing in smart grids. The importance of this issue is that the lack of access to / from the information could be a serious problems to the electricity supply.
- **Integrity:** Protection is against improper changes or destruction of data, in order not to deny the accuracy of information. Invalid modification or destruction of information is the lack of integration which can then cause disruption in power management decisions.
- **Availability:** Maintain a valid restrictions on access and disclosure of information, is primarily for specific information and privacy. This is to prevent invalid access to information which is not provided to the public and people [4].

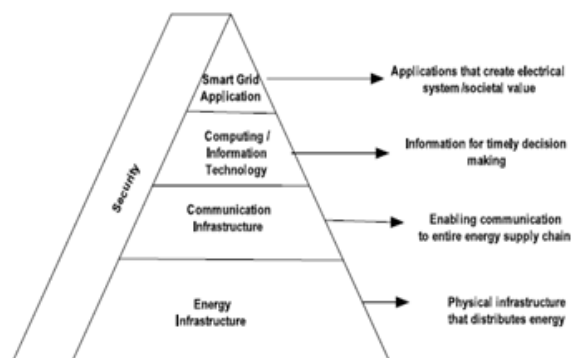


Figure 1. The concept of smart grid systems [3].

A. Security Requirements of the Smart Grid Power

Availability, integrity and confidentiality, are three main security objectives for smart grids. In addition to such a high level objectives, NIST report suggests specific security requirements for smart grids that are for both network security and physical security. Specifically, the network security specifies requirements for information to be accurate and intelligent network security and network systems; Smart grid cyber threats such as Stuxnet worm [5], has the potential to breach the national security, economic stability and even physical security. Power plants and SCADA systems (supervision and collecting data system) are always targeted by hackers. Moving from the close control systems to the open IP network, is associated with a range of vulnerabilities. For example, data integrity and authentication can be threatened by attacks such as spoofing or deny of service. Similarly, data security can expose to danger by built-destructive attacks such as viruses and Trojans [4]. Security in Smart grid is checked in both cyber and physical levels. Smart grid cyber security must a) Prevent the occurrence of such attacks and b) When the incidence of attacks (successful), has a mechanism which is restored and permanent.

The physical security, specifies the requirements for physical equipment and environmental protection as well as privacy security of staff and crew.

TABLE I. NETWORK SECURITY REQUIREMENTS IN SMART GRID ELECTRICITY COMPARED TO INTERNET SECURITY REQUIREMENTS [4]

Security functions	Smart Grid communication network	The internet
Authentication and access control	Strictly enforced for all communication flows throughout the system	Mostly free end-to-end without access control
Attack detection and countermeasures	Essential and widely-deployed everywhere	Mainly for critical routers and servers
Every node	Basic cryptographic functions	No specification
Security for network protocols	From MAC-layer to application-layer security	From network-layer to application-layer security

Since the security is the purpose of information and network systems, the security requirements for smart grids are summarized as follows:

- Identify the attack and reactionary measures
- Identification, authentication and access control
- Efficient and secure communication protocols

Table I, Specifies the network security requirements for Smart grid in comparison with Internet security requirements, briefly. In Table I, note that Smart Grid applies more serious security requirements to the Internet to ensure that a safe and efficient delivery of information happens for critical electricity infrastructure.

III. SMART GRID SECURITY THREATS

In communication networks, security attacks can be divided into two categories: Selfish members with abusive use and malicious users. Egoistic abusive users, are those to achieve more network resources than authorized users, by trying to do violation against communication protocols. In contrast, malicious users are not looking for personal gain, but the purpose of the acquisition, is modification or destruction of information in the network. Accordingly, both groups create challenging security issues for communication

networks. But in smart grid electricity, destructive behavior is more worrying issue than abusing, because millions of computing devices for surveillance purposes have been employed instead of providing data services such as downloading and file publication. Thus, malicious attacks can put significant catastrophe losses to power sources and a wide cut over for all power sources, which of course is prohibited in Smart Grid electricity. Since the numeration of the every possible attacks on smart grid is not possible due to extension, here malware attacks based security objectives are examined on three categories that of accessibility, lack of defects and confidentiality, as shown in Fig. 2-1.

- The attacks are targeted on availability, which are also called prohibited service attacks, are trying to put delay in smart grid communications or block or destroy them.
- Attacks which are targeted the unity factor, aimed at changing or destructing the data exchange deliberately for Smart Grid.
- Attacks that challenge the confidentiality, are aimed at receiving unauthorized information of network resources in Smart Grid.

IV. CRYPTOGRAPHY METHODS FOR SMART GRID

Methods based on cryptography have been transformed into countermeasures attacks on the integrity and reliability. Encryption is a basic cryptography method to protect data and make secure connection for any information systems. In a Smart Grid, the majority of electronic devices are expected to have the least basic encryption capabilities including the ability to support symmetric encryption [3].

Since the smart grid communication network consists of millions of computing systems with limited computing power, computational efficiency has become an important factor for adaptation encryption scheme with smart grid. Encryption schemes could be based on symmetric encryption (e.g. AES, DES) or asymmetric encryption (e.g. RSA). Symmetric encryption uses the same key for encryption and decryption. Asymmetric encryption uses public and private keys to decrypt and encrypt respectively. In general, asymmetric encryption requires more computing sources than symmetric encryption for a long key size (strong security). Therefore, the use of asymmetric encryption may be restricted just in computing systems.

Key management is also carried out in two ways which are elaborated below:

- Public Key Infrastructure: since the public-key algorithms are much slower than private key algorithms, these keys are used to distribute the session key (not encryption)
- Symmetric management: the best parts of public key infrastructure are the very low rate and their large key length. Although these two methods are complementary, the public key methods cannot be considered as alternative symmetric approaches. Smart meters have limited sources and weak computing power; so, according to traffic requirements, symmetric encryption algorithms are used to secure exchange messages [3],[6].

Researchers are trying to use an effective method to create a session key between smart meters and control center.

Moreover, symmetric encryption methods are mainly use for increasing efficiency and reducing overhead related to credits.

Recently, on the basis of this strategy, a symmetric key distribution method suggested by [7], that includes the both side connections and a reliable provider T, which shared a secret key for communicating parties for long period of time. The main purpose of this method is to create a dynamic session key between the two sides. The plan has been fully explained in the next section [7].

V. REVIW OF THE SKDP METHOD

In this plan, there are a trusted third party (TA), Smart meters which demand the connection (between (M) and a data aggregation server) communications responders (SP)). This method begins with smart meter M. M is asked for creating key session with the S server.

The process of exchanging messages come in Fig. 2.

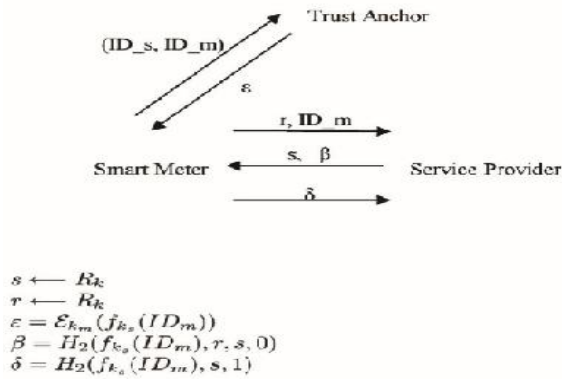


Figure 2. The flow of exchanged messages in the proposed method in [7]. Note: The signs used in this method are described in Appendix 1, Table II.

In [8] which the weaknesses of [7] was introduced, it was shown that the distribution of session key method is vulnerable against impersonate attack respondent (the attacker masquerade itself instead of the respondent at the beginning of the connection) (Fig. 3). In SKDP method to start session, initiator of M, sends to T the identity and respondent S such as (IDS, IDM). T calculate a sign with the use of the secret key S and it is encrypted with secret key M, although the attacker A cannot decode this sign, it may force T to produce the signs of A indicators instead of the calculating signs related to S. A does this by changing the initial message (IDS, IDM) to (IDA, IDM), so A can cause the M to get PRF (KA, IDM) Instead of PRF (KS, IDM) value. The important point is that M cannot be realized if the sign is made true or not. So A would put on itself instead S, because A is able to calculate token = Dec (KM, ε) = PRF (KA, IDM) and session key is calculated by M are SK = H (PRF (KA, IDM), s, IDM, IDS) that A owns all of its components and can calculate them as well.

VI. IMPROVED THE PROPOSED METHOD BY XIA AND WANG BASED ON THE RANDOM NONCE VALUE (NSKDP METHOD)

SKDP method is vulnerable against the impersonated respondent attack (the attacker masquerade himself instead of the respondent at the beginning of the connection). The first solution is to improve the security of the suggested method [7] against the impersonation attack, is based on a random value between T and M which is agreed as the session key.

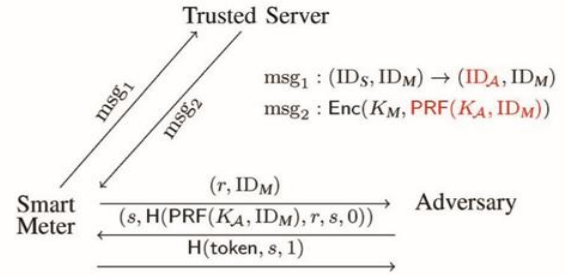


Figure 3. The impersonation attack in SKDP [8].

There was a problem that the transmitted message between M and T was clear text, there the attacker can easily change the amount of IDS and IDM, the first suggestion is to encrypt the values (IDS, IDM) with the KM value. In this case, only T and M can decrypt them. So the exchanged messages between the T and M are as follows. When Trusted server received SMS, then decode the message with the KM key. In this case, no one can see the meter and server identifiers in clear text. We named this method DSKDP.

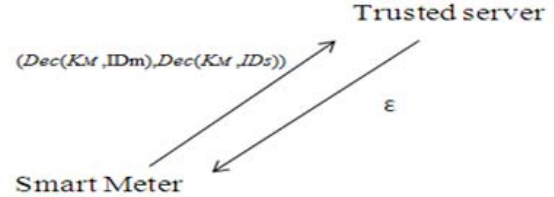


Figure 4. The flow of exchanged messages between the Trusted server and Smart Meter in DSKDP method.

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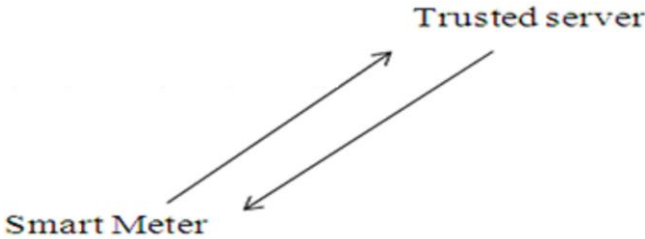
(PRF (KA, IDM), s, IDM, IDS) that A owns all of its components and can calculate them as well.

But this approach has a major problem: many of the Smart Grid nodes have limited sources and are not capable of doing encryption algorithms.

Since the encryption method may take a long time, it is recommended to use a Nonce value to increase the security. It is suggested to create a session key in a very short possible time, and the value recognizer XOR with session key and then sent to the trusted server. H is a one-way hash function. The way it works is as follows:

- The meter selects a nonce value, called NM and sends the req (IDM, NM $\oplus$ kM) message to the Trusted server.
- Trusted server selects the NT and the NM's value ($NM \oplus kM \oplus kM$) is obtained and sends the ((Chall ($NT \oplus h(KM \parallel NM)$, $h(KM \parallel NT \parallel NM)$)) message to the meter.
- The meter calculates the $h(KM \parallel NM \parallel NM)$ value, if it is equal to the value which T has sent the server is valid, otherwise it will deny the connection. And if the server was valid, then sends the Res (IDM, $h(IDM, h(NT \parallel kM \parallel NM))$) message to the T and the NT's value will be the session key.
- T obtains its information and gets the second part IDM's value from the meter and the meter will be valid if they are equal. Otherwise it will drop the connection and prevent DOS attacks. NT will be the session key if it is valid.

After the session has made up and agreed on the key and two-way authentication, the meter can send (IDM $\oplus$ NT, IDS $\oplus$ NT) message to the T.



Messages:

- 1-req (IDM, NM $\oplus$ kM)
- 2-chall($NT \oplus h(KM \parallel NM)$, $h(KM \parallel NT \parallel NM)$)
- 3- Res(IDM,, $h(IDM,, h(NT \parallel kM \parallel NM))$)

Figure 5. The flow of exchanged messages between the trusted server and smart meter for creating session key between themselves in NSKAD method.

VII. ANALYSIS OF PROPOSED METHOD

Availability, integrity and confidentiality are three main objectives on the Smart Grids security. The proposed method is a suitable approach against attacks that challenge confidentiality, and aim to get unauthorized information from the network sources in the Smart Grid.

Our method is based on the Nonce value and XOR function and does not use any difficult mathematics. Since a simple function such as the hash function is used, proposed

method is considered as low overhead and therefore is suitable for Smart meters with limited computing sources. Note that NONCE is changed for each session. The message which is swapped between smart meters and reliable server is not as clear text. So the attacker cannot spoof the initial message and make the server instead of meter to create symptoms. Create a light session key between the trusted server and meter, prevents the impersonation attack. The key is changed each session so the information confidentiality is maintained. In step 3 the server is being authenticated, and if an invalid server tend to impersonate the trusted server, by examining the $h(KM \parallel NT \parallel NM)$ message the connection will be disconnected immediately.

Furthermore, based on the following evaluation of performance in, we can see our scheme performances more efficiently Fig. 6.

According to Fig. 6, the runtime of NSKDP method is less than SKDP, as it is suitable for the close connection to real-time meters and servers which accumulate data.

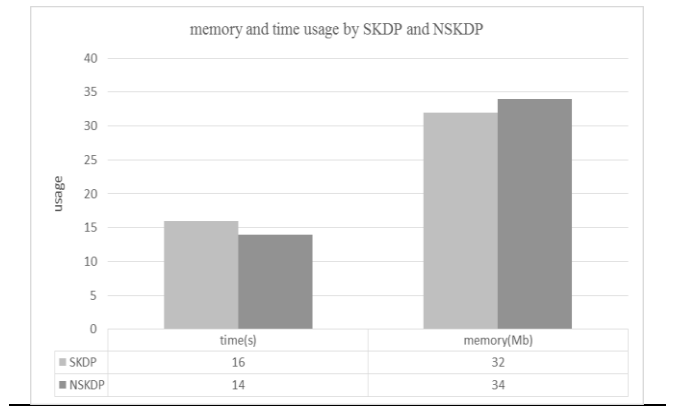


Figure 6. Efficiency comparidion of NSKDP and SKDP.

TABLE II. NOTATION SUMMARY

1- trusted server selects two prime numbers P and P from $p = q + 1$ and a random number as its private key from $\mathbb{Z}^* \times \mathbb{Z}^*$
2- $ID_M \in \{0,1\}^*$ and $ID_S \in \{0,1\}^*$ are server and meter identifiers
3- D (.) and E (.) are the public key encryption algorithms such as (AES, etc.)
4- H2 (.) and H1 (.) are one-way hash functions
5- $K_i = H1 (ID_{x_i} \text{ mod } p)$ is the private key related to ID_i
6- For the meters, keys must be stored in small temporary memory.
7- $r \leftarrow R_k(.)$, K map the random bits to r
8- PRF use a pseudo-random generator function for K parameter which is generated a K-bit key and a K-bit output

VIII. CONCLUSION

Environmental and economic factors have led the old electrical infrastructure became unstable; Thus, the need for reliability, compatibility, security and cost-effectiveness was felt. Consequently, Smart Grid is the result of two important trends, digitization and access to wireless and wired technologies, is as significant development in telecommunications and information technologies. These networks use the power streams and the two-way data network, to provide automated power distribution widely. One of the distinctive measures in smart network

connections is vast expansion of sensors and Smart meters. Thus, there will be a large amount of data, meters information, sensors, monitoring.

Since there are different kinds of systems in Smart Grids and these systems need to interact and cooperate with each other as coordinated, in order not to occur an interference in their work, security and privacy play an important role in Smart Grids and smart meter networks. Challenges of smart meters are more dependent on communication technologies. The main challenges of Smart meters are sources restrictions, computational bandwidth limitations, memory constraints. In the proposed method, sending the server and meter's information in clear text is prevented by a Nonce, hash functions and XOR. The benefits of the proposed method is a two-way authentication is done with minimal computing, and easily prevents the impersonation of both sides.

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